Learning a network digital twin as a hybrid system^{☆,☆☆}Christos Mavridis^a, Fernando S. Barbosa^b, Hamed Farhadi^b, Karl H. Johansson^a^a Division of Decision and Control Systems, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden^b Ericsson Research, Stockholm, Sweden

ARTICLE INFO

Keywords:

Control under communication constraints
 Hybrid and switched systems modeling
 Learning methods for control
 Machine and deep learning for system
 identification

ABSTRACT

Network digital twin (NDT) models are virtual models that replicate the behavior of physical communication networks and are considered a key technology component to enable novel features and capabilities in future 6G networks. In this work, we focus on communication-aware control applications and study NDTs that model the communication quality properties of a multi-cell, dynamically changing wireless network over a workspace populated with multiple moving users. We propose an NDT modeled as a hybrid system, where each mode corresponds to a different base station and comprises sub-modes that correspond to areas of the workspace with similar network characteristics. The proposed hybrid NDT is identified and continuously improved through an annealing optimization-based learning algorithm, driven by online data measurements collected by the users. The advantages of the proposed hybrid NDT are studied with respect to memory and computational efficiency, data consumption, and the ability to timely adapt to network changes. Finally, we validate the proposed methodology on real experimental data collected from a two-cell 5G testbed.

1. Introduction

Network digital twins (NDTs) are virtual models that replicate the behavior and structure of physical communication networks, facilitating real-time visualization, simulation, and performance analysis and optimization [1]. These are treated as one of the key technologies towards the 6G vision of connecting the physical and digital worlds [2]. These digital counterparts allow network administrators to evaluate operational metrics, foresee potential issues, and experiment with multiple configurations in a risk-free environment, thereby minimizing service disruptions [3]. This approach underpins key advancements such as automated network management, predictive upkeep, and efficient resource deployment, contributing to improved robustness and adaptability of communication networks [4]. Given the increasing complexity of communication networks and the advancements in computational hardware enabling the learning of complex systems [5], NDTs are anticipated to become essential tools for managing the intricate dynamics of modern communication infrastructures and ensuring the maintenance of high-quality service levels. Consequently, NDTs are expected to play a critical role in future 6G networks [6].

In this work, we focus on NDTs that model the communication quality properties of a multi-cell, dynamically changing wireless network over a workspace populated with multiple, moving user equipments (UEs). These types of NDTs have wide range of applications, including resource allocation [7] and communication-aware motion planning, where the objective is to design optimal

[☆] This article is part of a Special issue entitled: 'TC 1.5 Networked systems (IFAC WC 2026)' published in Nonlinear Analysis: Hybrid Systems.

^{☆☆} Research partially supported by the Swedish Foundation for Strategic Research (SSF) grant IPD23-0019.

* Corresponding author.

E-mail addresses: mavridis@kth.se (C. Mavridis), fernando.dos.santos.barbosa@ericsson.com (F.S. Barbosa), hamed.farhadi@ericsson.com (H. Farhadi), kallej@kth.se (K.H. Johansson).

<https://doi.org/10.1016/j.nahs.2026.101746>

Received 31 October 2025; Received in revised form 6 May 2026; Accepted 14 May 2026

Available online 20 May 2026

1751-570X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

decisions for the UEs while maintaining and optimizing the connectivity to a base station [8,9]. Regarding the network properties, most of the works in the literature only considered the signal strength of each UE, ignoring terms related to the Quality of Service (QoS), such as throughput, end-to-end latency affected by routing, or the impact of the UE handover in the presence of multiple base stations [8–11]. In particular, considering scenarios such as high-band transmissions in densely populated areas with multiple base stations, handover modeling becomes a critical component of a wireless network that need to be well considered in the NDT design and operation.

In addition to the limitation in the representation of network properties, full knowledge of the wireless channel spatial variations is a common assumption in the literature [9,11,12]. In the absence of this assumption, learning-based [13] and data-driven methods can be employed to identify the NDT [14,15]. Most works make use of a probability density model to estimate the network properties [10,15,16], while others use model-based learning methods [17]. However, accurately reconstructing a network model completely from data can be challenging, and sensory measurements can be slow and expensive to acquire [14,18–21]. Thus, designing a data-based, explainable, resource-efficient NDT model with fast adaptation mechanisms still remains an open problem with important implications.

1.1. Contribution

Towards this direction, we propose the modeling and identification of an NDT model as a hybrid system that (a) has multiple modes, each corresponding to one of potentially many base stations of the network, and (b) is identified and continuously updated through a data-driven and explainable prototype-based learning algorithm based on principles from homotopy optimization [22,23].

To the best of our knowledge, the advantages of modeling an NDT as a hybrid system with different operating modes have not been studied before. In this work, we show that a hybrid NDT can facilitate not only a more accurate representation of the wireless network, but also the ability to adapt to rapid network changes faster and more efficiently in terms of computation and data usage.

We develop a hybrid identification algorithm as an extension of a prototype-based learning method trained using principles from online deterministic annealing (ODA), a homotopy optimization approach, first introduced for clustering problems [23,24]. ODA makes use of entropy regularization to progressively estimate the number of clusters needed and provide real-time control of the performance-complexity trade-off. Based on this approach, we construct a combined clustering, classification, and regression algorithm for the identification of an NDT as a hybrid system.

Finally, we compare our methodology against a standard identification algorithm in a 5G network with two base stations and event-triggered network changes located at Ericsson's headquarters, Stockholm, Sweden.

1.2. Notation

The sets $\mathbb{R}$ and $\mathbb{R}_+$ represent the sets of real and non-negative real numbers, respectively. Unless otherwise specified, random variables $\mathcal{X} : \Omega \rightarrow \mathbb{R}^d$ are defined in a probability space $(\Omega, \mathbb{F}, \mathbb{P})$. The probability of an event is denoted $\mathbb{P}[\mathcal{X} \in S] := \mathbb{P}[\omega \in \Omega : \mathcal{X}(\omega) \in S]$, and the expectation operator $\mathbb{E}[\mathcal{X}] = \int_{\Omega} \mathcal{X} d\mathbb{P}$. In case of multiple random variables $(\mathcal{X}, \mathcal{Y})$ and a deterministic function f , the expectation operator $\mathbb{E}[f(\mathcal{X}, \mathcal{Y})]$ is understood with respect to the joint probability measure, while $\mathbb{E}[\mathcal{X}|\mathcal{Y}] := \mathbb{E}[\mathcal{X}|\sigma(\mathcal{Y})]$ denotes the expectation of $\mathcal{X}$ conditioned to the σ -field of $\mathcal{Y}$. Stochastic processes $\{\mathcal{X}(k)\}_k$, $k \in \mathbb{Z}_+$, are defined in the filtered probability space $(\Omega, \mathbb{F}, \{\mathcal{F}_n\}_n, \mathbb{P})$, where $\mathcal{F}_n = \sigma(\mathcal{X}(k)|k \leq n)$, $k \in \mathbb{Z}_+$, is the natural filtration. The indicator function of the event $[\mathcal{X} \in S]$ is denoted $\mathbb{1}_{[\mathcal{X} \in S]}$.

2. Hybrid network digital twin model

In this work, we are interested in NDT representations that can model both continuous-valued network properties, such as signal strength and latency, and discrete-valued properties, such as cellular association between base stations. In this section, we define the properties of a general NDT representation and develop an NDT with hybrid structure as a modeling design that enables modular representation and fast adaptation.

2.1. Network properties

Consider a workspace $W \subset \mathbb{R}^d$ containing n moving agents at locations $p_i(t) \in W$, $i \in \{1, \dots, n\}$, $t \in \mathbb{R}_+$, and m base stations $\{I_1, \dots, I_m\}$, at fixed positions $q_i \in W$, $i \in \{1, \dots, m\}$. The workspace W usually represents a two-dimensional ($d = 2$) or three-dimensional ($d = 3$) space but can be generalized to more dimensions. A representation of a 2D workspace is depicted in Fig. 1.

Within the workspace W , electromagnetic signals are transmitted carrying information communicated through the network. The network is responsible for immediate and fast connectivity and maximization of the communication quality of the users. Among the various communication quality metrics that can be measured in a wireless 5G/6G network, the most important in communication-aware cyber-physical systems and IoT applications are signal strength, user latency, channel throughput, and communication delay due to handovers between cells [25].

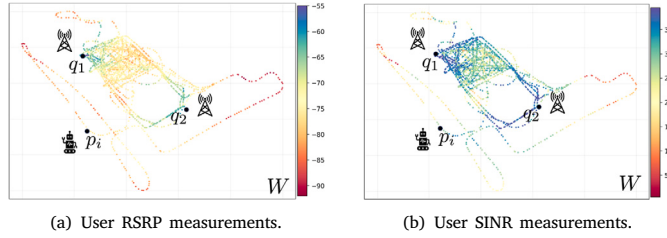


Fig. 1. Illustration of a 2D workspace with two cells and one moving UE providing network measurements.

2.1.1. Signal strength

Signal strength comprises mainly three components: path loss (low-frequency decay in the channel power with distance), shadowing (high-frequency fading mainly due to obstructing structures), and multi-path fading (very high-frequency attenuation due to scattering and reflection), given the locations of the base stations [8,9]. Signal strength can be measured by numerous metrics, including the Reference Signal Received Power (RSRP) and the Signal-to-Interference-plus-Noise Ratio (SINR) observed by the user [26].

2.1.2. Latency and channel throughput

Throughput represents the actual rate at which user data is successfully transmitted over a wireless channel, and depends on SINR, allocated bandwidth, and the radio access technology (e.g. code rate, modulation, and waveform). Latency is time delay experienced for a data packet to travel from the source to the destination across the network, including transmission delay, processing delay, propagation delay, and queuing delay. Latency is also affected by SINR, network congestion, and radio access technology (e.g. coding, modulation, beamforming, ...) [26].

2.1.3. Handover delays

A handover is the process of transferring an active connection (voice or data) from one base station to another to maintain service continuity as the user moves or radio conditions change. A handover is triggered when the UE detects that the signal quality (e.g., RSRP or SINR) from a neighboring cell becomes better than that of the serving cell over a time window, or when network conditions require transferring the connection to maintain service quality, or based on the predictions of the signal quality [27]. Handovers can increase packet latency and delay due to the time required for signaling, context transfer, and resource reallocation between cells during the connection switch.

2.2. Network digital twin

Given the wireless network properties of interest in a workspace W , we define a Network Digital Twin model as follows:

Definition 1 (Network Digital Twin). Let $t \in \mathbb{R}_+$ represent time, and $x \in W \subset \mathbb{R}^d$ be a position vector in the workspace. Define $Q(t, x) \in \mathbb{R}^l$ a vector function of l scalar quantities measuring the communication quality (such as signal strength, latency, or throughput), and $c(t, x) \in \{1, 2, \dots, m\}$ a discrete-valued function representing the index of the ID of the cell connected to an agent at position x , such that $I = \{I_1, I_2, \dots, I_m\}$. Then, a Network Digital Twin (NDT) model $\tilde{N}$ is defined as a function $N : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^{l+1}$, that maps the time and position vector to a vector of communication quality and cell ID index, i.e.,

$$\tilde{N} : (t, x) \mapsto (Q, c). \quad (1)$$

To validate or identify an NDT model of the form (1), observations O_i can be collected from each agent $i \in \{1, 2, \dots, n\}$, such that:

$$O_i(t, x) = \begin{bmatrix} t \\ x(t) \\ Q(t, x) \\ c(t, x) \end{bmatrix}, \quad (2)$$

where, the components of $Q(t, x) \in \mathbb{R}^l$ are quantified by easily obtainable network measurements, including user RSRP, user SINR, communication latency or throughput, etc. Thus, the observation vector consists of easily obtained measurements and gives sufficient information that best represents the behavior of the network with respect to model (1).

2.3. Network digital twin as a hybrid system

In this section, we propose the representation of the NDT in (1) as a hybrid system. The hybrid structure offers a modular and explainable representation of the network properties, simplifies the learning models used in data-driven approaches, and allows quick adaptation to rapid changes that occur to a base station, such as cell malfunction or overload.

A general input–output representation of a hybrid system can be written in the form:

$$\begin{cases} y(t) = N_\sigma(t, x) \\ \sigma(t) = \phi(t, x), \end{cases} \quad t \in \mathbb{R}_+, \quad (3)$$

where $x \in W \subset \mathbb{R}^d$ represents the input vector, $y(t) \in \mathbb{R}^l$ represents the output vector, and $\sigma(t) \in \{1, \dots, s\}$ is called the mode-switching signal and can take a finite number of s values that are called the modes of the hybrid system. The functions $N_i : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^l$, $\forall i \in \{1, \dots, s\}$, and $\phi : \mathbb{R}^{d+1} \times \mathbb{R}^d \rightarrow \{1, \dots, s\}$ define the continuous and discrete dynamics of the hybrid system, respectively.

To capture cell-specific properties and handover phenomena, we focus on a special case of hybrid systems, namely switched systems for which ϕ is given by:

$$\phi(t, x) = c(t, x), \quad t \in \mathbb{R}_+, \quad (4)$$

such that the NDT has m modes, each one describing the network properties when connected with each base station with ID I_i and position q_i , $i = 1, \dots, m$. This modularity provides the necessary information to detect and predict handovers, as well as a framework to detect and update the NDT model when network changes depend on cell-specific network events, such as cell malfunction or overload.

While (3), (4) provide an intuitive hybrid structure for the NDT, in practice, the rule $\sigma(t) = c(t, x)$ defines a time-dependent mode switching behavior that is difficult to compute, represent with a model, or estimate from observations. Therefore, we propose an alternative representation. We start with a partition-based hybrid representation of the form:

$$\sigma(t) = i \Leftrightarrow x \in S_i(t), \quad t \in \mathbb{R}_+, \quad (5)$$

where the set $\{S_i(t)\}_{i=1}^s$ defines a partition of W , i.e., $S_i(t) \cap S_j(t) = \emptyset$ for $i \neq j$, and $\bigcup_i S_i(t) = W$, $\forall t \in \mathbb{R}_+$. It is easy to see that the representation (5) is equivalent to (4) for a time-dependent partition $\{S_i(t)\}_{i=1}^s$ such that $x \in S_i(t) \Leftrightarrow c(t, x) = i$, $\forall t \in \mathbb{R}_+$.

To simplify the modeling and learning complexions of this representation, we need to (a) address the time dependency, and (b) find a simple representation of the non-convex regions $S_i(t)$, $i = 1, \dots, s$. Time dependency will be discussed in Sections 3.1 and 4. Regarding the representation of the modes, we introduce a finer polyhedral partition $\{\Sigma_i(t)\}_{i=1}^K$ of W , for some $K > s$, such that

$$S_i(t) = \bigcup_{j \in J_i} \Sigma_j(t), \quad i = 1, \dots, s, \quad t \in \mathbb{R}_+, \quad (6)$$

with $J_i \cap J_j = \emptyset$ and $\bigcup_i J_i = K$. In other words, each $S_i(t)$ can be approximated by the union of one or more $\Sigma_j(t)$ at any time instance $t \geq 0$. Finally we also introduce the variables $\{\bar{c}_i\}_{i=1}^K$ such that

$$\bar{c}_i = i \Leftrightarrow j \in J_i. \quad (7)$$

Now the hybrid NDT model can be written in the form:

$$\begin{cases} y(t) = N_\sigma(t, x) \\ \sigma(t) = \sum_j \mathbb{1}_{[x \in \Sigma_j(t)]} \bar{c}_j, \end{cases} \quad t \in \mathbb{R}_+. \quad (8)$$

A depiction of the model parameters of the NDT representation (8) is shown in Fig. 2. In the next section, we will make use of this NDT representation to construct a data-driven learning algorithm to identify and adapt the hybrid NDT.

3. Data-based hybrid NDT identification

In this section we will use a prototype learning method to represent the partition $\{\Sigma_i(t)\}_{i=1}^K$, estimate the variables $\{\bar{c}_i\}_{i=1}^K$, and approximate the models $\{N_i(t, x)\}_{i=1}^s$ of the hybrid NDT model (3).

3.1. Prototype-based learning model

To efficiently represent $\{\Sigma_i(t)\}_{i=1}^K$, we make use of a polyhedral Voronoi partition produced by a set of variables $\{\rho_i(t)\}_{i=1}^K$, $\rho_i(t) \in W$, $\forall t \in \mathbb{R}_+$, according to the rule:

$$\Sigma_i(t) = \left\{ x \in W : i = \underset{j}{\operatorname{argmin}} d(x, \rho_j(t)) \right\}, \quad t \in \mathbb{R}_+, \quad (9)$$

for a given dissimilarity measure d . For the Voronoi regions Σ_i to be polyhedral, we select d to be a Bregman divergence [23], a family of dissimilarity measures that includes, among others, the squared (weighted) Euclidean distance, and the Kullback–Leibler divergence [24,28]. To minimize the updates and data consumption of the learning process, we assume that $\rho_i(t)$ only change

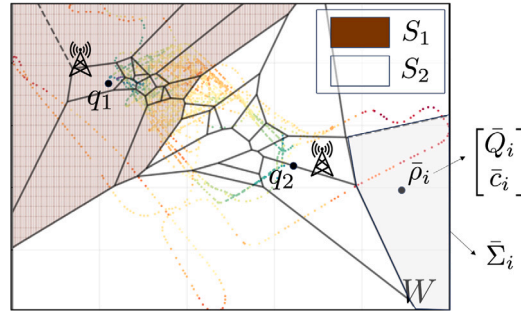


Fig. 2. Prototype-based representation of the Hybrid NDT.

during discrete events (e.g., cell overload or major network changes and workspace alterations) and model the dynamics of $\rho_i(t)$ as a first-order filter of the form:

$$\dot{\rho}_i(t) = -\gamma_\rho \rho_i(t) + \bar{\rho}_i^{(k)}, \quad \rho(0) = \rho_0, \quad (10)$$

for all $i \in \{1, \dots, K\}$, some $\gamma_\rho > 0$, and $t \in \mathbb{R}_+$. The parameters $\bar{\rho}_i^{(k)}$ are event-triggered functions that change in discrete time instances k and define the equilibrium of (10). In that way, the learning algorithm needs to estimate the optimal number and locations of the constant parameter vectors $\{\bar{\rho}_i\}_{i=1}^K$, as well as the association parameters $\{\bar{c}_i\}_{i=1}^K$, detect possible changes through an appropriate criterion, and provide a fast adaptation mechanism.

In addition to the above problem, which can be viewed as a combined clustering and classification problem, the estimation of $\{N_i(t, x)\}_{i=1}^s$ becomes an additional regression problem. By introducing a set of constant models $\{\bar{Q}_i(x)\}_{i=1}^K$ as local models for each region Σ_i , we model the time dynamics of each $N_i(t, x)$ as:

$$\begin{aligned} \frac{\partial}{\partial t} N_i(t, x) &= -\gamma_N N_i(t, x) + \sum_{j=1}^K \mathbb{1}_{[x \in \Sigma_j]} \bar{Q}_j^{(k)}(x) + \Delta_i^{(k)}(t), \\ N_i(0, x) &= N_i^0(x), \quad i \in \{1, \dots, s\}, \quad x \in W, \quad t \in \mathbb{R}_+, \end{aligned} \quad (11)$$

for some $\gamma_N > 0$. Model (11) is a first-order filter following the local models $\{\bar{Q}_i^{(k)}(x)\}_{i=1}^K$. The term $\Delta_i^{(k)}(t)$ models time-dependent changes that affect the operation of cell I_i , such as cell overload or malfunction. It is an adaptation mechanism that can only be exploited through the use of a hybrid system of the form (8). These time-dependent terms will be discussed in Section 4.

While there are many candidates for modeling $\bar{Q}_i(x)$, such as linear or quadratic functions, artificial neural networks, or Gaussian process regression models, we further compress the learning process by assuming constant models, i.e.,

$$\bar{Q}_i(x) = \bar{Q}_i, \quad \forall x \in W, \quad \forall i = 1, \dots, K. \quad (12)$$

Since $\bar{Q}_i(x)$ are local models within the polyhedral regions $\Sigma_i(t)$, spatial information is already stored in the shape, position, and area of $\Sigma_i(t)$. The optimization problem is thus transformed from a model regression problem to a partition optimization such that local constant models provide a good approximation of the spatial characteristics of the NDT model. As a result, the hybrid NDT model identification problem can be written as follows:

Problem 1 (Hybrid NDT Identification). Estimate the number K and the parameters $\{\bar{\rho}_i\}_{i=1}^K$, $\{\bar{c}_i\}_{i=1}^K$, and $\{\bar{Q}_i\}_{i=1}^K$, such that the partition $\{\Sigma_i(t)\}_{i=1}^K$ minimizes both error terms $d(\sigma(t), c(t, x))$ and $d(y(t), Q(t, x))$ for a given Bregman divergence d , as described above.

Problem 1 can be approached as a combined clustering and classification problem of the form:

$$\begin{aligned} \text{minimize } \mathbb{E} \left[\sum_{i=1}^K \mathbb{1}_{[x \in \Sigma_i]} d_c(Z, c, \mu_i, \bar{c}_i) \right] \\ \text{s.t. } \bar{\Sigma}_i = \left\{ x \in W : i = \underset{j}{\operatorname{argmin}} d(x, \bar{\rho}_j) \right\}, \end{aligned} \quad (13)$$

where the random observation vector:

$$Z = \begin{bmatrix} x(t) \\ Q(t, x) \end{bmatrix} \in \bar{W} \subseteq \mathbb{R}^{d+l} \quad (14)$$

is defined in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, c is a random variable following the distribution of $c(t, x)$, μ_i is the augmented codevector:

$$\mu_i := \begin{bmatrix} \bar{\rho}_i \\ \bar{Q}_i \end{bmatrix} \in \bar{W}, \quad i = 1, \dots, K, \quad (15)$$

and d_c is a dissimilarity measure of the form:

$$d^c(x, c_x, y, c_y) = \begin{cases} d(x, y), & c_x = c_y \\ \infty, & c_x \neq c_y \end{cases} \quad (16)$$

motivated by the combined clustering and classification problem formulation in [22,28].

Problem (13) is a hard clustering problem with respect to the parameters $\{\bar{\rho}_i, \bar{Q}_i, \bar{c}_i\}_{i=1}^K$ with known K . In Section 3.2, we make use of the Online Deterministic Annealing (ODA) method as a homotopy optimization approach to progressively estimate the optimal parameters and the minimum number K needed using an online, gradient-free recursion.

3.2. Online deterministic annealing

The Online Deterministic Annealing method is a recursive stochastic optimization algorithm that can solve problems of the form (13), by progressively estimating the number K of the augmented codevectors $\{\mu_i\}_{i=1}^K$ [24,28].

First a quantizer (encoder) $E: \bar{W} \rightarrow \bar{W}$ is defined as a discrete random variable in the same probability space with countably infinite domain $\mu := \{\mu_i\}$. Then the following multi-objective optimization problem is formulated

$$\underset{\mu}{\text{minimize}} F_\lambda(\mu) := (1 - \lambda)D(\mu) - \lambda H(\mu), \quad \lambda \in [0, 1), \quad (17)$$

where the term

$$D(\mu) := \mathbb{E}[d(Z, E)] = \int p(z) \sum_i p(\mu_i|z) d(z, \mu_i) dz$$

is a generalization of the objective in (13), and

$$\begin{aligned} H(\mu) &:= \mathbb{E}[-\log P(Z, E)] \\ &= H(Z) + H(E|Z) \\ &= H(Z) - \mathbb{E}[\log p(E|Z)] \\ &= H(Z) - \int p(z) \sum_i p(\mu_i|z) \log p(\mu_i|z) dz \end{aligned}$$

is the Shannon entropy. This is now a problem of finding the locations $\{\mu_i\}$ and the corresponding probabilities $\{p(\mu_i|z)\} := \{p(E = \mu_i|Z = z)\}$. The Lagrange multiplier $\lambda \in [0, 1)$ controls the trade-off between D and H . The entropy term, introduces several properties to the approach including robustness with respect to initial conditions [28,29]. In addition, reducing the values of λ defines a direction that resembles an annealing process [28] and a bifurcation phenomenon, according to which the number K of the codevectors increases as the value of λ decreases below certain data-dependent values [22,24].

As shown in [24], a sufficient solution to (17), for a fixed value of λ , is given by

$$\mu_i^* = \mathbb{E}[Z|\mu_i] = \frac{\int z p(z) p^*(\mu_i|z) dz}{p^*(\mu_i)}, \quad (18)$$

where

$$p^*(\mu_i|z) = \frac{e^{-\frac{1-\lambda}{\lambda} d(z, \mu_i)}}{\sum_j e^{-\frac{1-\lambda}{\lambda} d(z, \mu_j)}}, \quad \forall z \in S, \quad (19)$$

as long as d is a Bregman divergence. In addition, the following Lemma constructs a gradient-free stochastic approximation algorithm for recursive estimation:

Lemma 1 (28). *The sequence $\mu_i(\tau)$ constructed by the recursive updates*

$$\begin{cases} \pi_i(\tau+1) &= \pi_i(\tau) + \beta(\tau) [s_i \hat{p}(\mu_i|z_\tau) - \pi_i(\tau)] \\ \sigma_i(\tau+1) &= \sigma_i(\tau) + \beta(\tau) [s_i z_\tau \hat{p}(\mu_i|z_\tau) - \sigma_i(\tau)] \end{cases} \quad (20)$$

where $z_\tau \sim Z$, $s_i := \mathbb{1}_{[\bar{c}_i=c]}$, $\sum_\tau \beta(\tau) = \infty$, $\sum_\tau \beta^2(\tau) < \infty$, and the quantities $\hat{p}(\mu_i|z_\tau)$ and $\mu_i(\tau)$ are recursively updated as follows:

$$\mu_i(\tau) = \frac{\sigma_i(\tau)}{\pi_i(\tau)}, \quad \hat{p}(\mu_i|z_\tau) = \frac{\pi_i(\tau) e^{-\frac{1-\lambda}{\lambda} d(z_\tau, \mu_i(\tau))}}{\sum_i \pi_i(\tau) e^{-\frac{1-\lambda}{\lambda} d(z_\tau, \mu_i(\tau))}}, \quad (21)$$

converges almost surely to a solution of (18).

Lemma 1 describes how to solve the optimization problem for a given value of the parameter λ . In the online deterministic annealing approach, a sequence of optimization problems with decreasing values of λ is solved, forming a homotopy optimization method. It is shown that the unique values of the set $\{\mu_i\}$ that solves (17), form a finite set of $K(\lambda)$ values, referred to as “effective codevectors”. At high values ($\lambda \rightarrow 1$), (19) yields uniform association probabilities $p(\mu_i|z) = p(\mu_j|z)$, $\forall i, j, \forall z$, and as a result of (18), all pseudo-inputs are located at the same point $\mu_i = \mathbb{E}[Z]$, $\forall i$ which means that there is one unique “effective” codevector given

by $\mathbb{E}[X]$. As λ is lowered below a critical value, a bifurcation phenomenon occurs, when the number of “effective” codevectors increases, which describes an annealing process [22,28]. Mathematically, this occurs when the existing solution μ^* given by (18) is no longer the minimum of (17), as the temperature λ crosses a critical value.

In other words, the number of codevectors increases countably many times as the value of λ decreases, and an algorithmic implementation needs only as many codevectors in memory as the number of “effective” codevectors. In practice, the bifurcation points are detected by introducing perturbing pairs of effective codevectors at each temperature level λ . In the case of classification, we represent the set of unique class labels as $\underline{c}$. Then a perturbed copy of each effective codevector is produced for each unique known class label $\underline{c}_j \in \underline{c}$ as follows:

$$\begin{aligned}\bar{\mu} &\leftarrow \bigcup_{\bar{\mu}_i \in \bar{\mu}} \{\bar{\mu}_i + \delta_j\}_{j \in \{1, \dots, |\underline{c}|\}} \\ \bar{c} &\leftarrow \bigcup_{i \in \{1, \dots, |\bar{\mu}|\}} \{\bar{c}\}_{\bar{c} \in \underline{c}}.\end{aligned}\quad (22)$$

where $|\underline{c}|$, $|\bar{\mu}|$ represents the cardinality of the sets $\underline{c}$, $\bar{\mu}$, respectively. The newly inserted codevectors will merge with their pair if a critical temperature has not been reached and separate otherwise, reflecting a genetic-type approach [24]. For a fixed temperature coefficient λ the complexity of the algorithm is

$$O(N_c(2K)^2 d_Z), \quad (23)$$

where N_c is the number of stochastic approximation iterations needed for convergence, K is the number of codevectors, and d_Z is the dimension of the input vectors Z . A detailed discussion on the implementation of the original online deterministic annealing algorithm, its complexity, and the effect of its parameters can be found in [22,24,28]. An important parameter that can influence the final NDT model representation is the stopping criterion (final value of λ), which, in this work, is selected by computational and training error thresholds.

4. Time dependency

The learning algorithm proposed in Section 3 estimates the parameters $\{\bar{\mu}_j, \bar{Q}_j, \bar{c}_j\}_{j=1}^K$, along with the number K . These parameters are estimated in discrete time, according to the frequency and availability of the observations. Eqs. (10) and (11) constitute filtering equations for a stable continuous-time approximation of the hybrid NDT. In this section, we focus on the design of event-trigger mechanisms to detect changes in the values $\{\bar{\mu}_j, \bar{Q}_j, \bar{c}_j\}_{j=1}^K$, which, by construction, represent local averages in the region $\bar{\Sigma}_i$. Therefore, a change in these local average values signifies a time-dependent drift in the model, which triggers a new training cycle and/or corrections terms, such as the quantities $\Delta_i^{(k)}(t)$ in (11).

4.1. Regression error event trigger

We detect a communication quality regression error event by the condition:

$$\|Q(t, x) - \sum_{j=1}^K \mathbb{1}_{[x \in \Sigma_j]} \bar{Q}_j\| \geq \epsilon_q \quad (24)$$

for some $\epsilon_q > 0$. This type of drift can signify local or global network changes. Therefore, in addition to the parameters $\{\bar{Q}_i\}$, the partition $\{\bar{\Sigma}_i\}$ needs to be updated as well. The drift detection triggers the algorithm to increase the parameter λ by a predefined amount, and continue the stochastic approximation updates of the learning algorithm. Increasing the parameter λ naturally resets the algorithm to focus on larger changes. The parameter λ decreases back to its original value according to the algorithm, having adjusted to the changes without the need to learn the model from scratch.

4.2. Classification error event trigger

We detect a cell ID mode classification error event by the condition:

$$\sum_i \|c(t, x_i) - \sum_{j=1}^K \mathbb{1}_{[x_i \in \Sigma_j]} \bar{c}_j\| \geq \epsilon_s, \quad (25)$$

for all x_i in a rolling window of observations and some $\epsilon_s > 0$. Similar to the regression error, the misclassification detection triggers the algorithm to increase the parameter λ by a predefined amount, and continue the stochastic approximation updates of the learning algorithm, avoiding the need to learn the model from scratch.



Fig. 3. Experimental apparatus at Ericsson's headquarters, Stockholm, Sweden.

4.3. Cell-specific error event trigger

In a 6G network with multiple base stations, it is often the case that the communication quality between a base station I_i , $i \in \{1, \dots, m\}$, and the UEs connected to them is affected due to a cell malfunction or high user demand. In this case, the user experiences high latency that is often reflected in the SINR metric. A condition can be constructed if a latency or SINR component $Q_\tau(t, x)$, $\tau \in \{1, \dots, l\}$, is observed by the network, such that:

$$\|Q_\tau(t, x) - \sum_{j=1}^K \mathbb{1}_{[x \in \Sigma_j]} \bar{Q}_{j_\tau}\| \geq \epsilon_\tau \quad (26)$$

for some $\epsilon_\tau > 0$. This induces a drift in the hybrid NDT model, that is shared among the mode $\sigma(t) = i$, and is usually temporary. In that case, the correction term $\Delta_i^{(k)}(t)$ in (11) is used, defined as follows:

$$\Delta_i^{(k)}(t) = \begin{cases} Q(t, x) - \sum_{j=1}^K \mathbb{1}_{[x \in \Sigma_j]} \bar{Q}_{j_\tau}, & t_k < t < T \\ 0, & t > t_k + T \end{cases} \quad (27)$$

where t_k represents the time of the event trigger (k th discrete timestep), and $T > 0$ is a predefined time window. The term $\Delta_i^{(k)}(t)$ fades after time T , and no re-training is necessary.

5. Experimental results and discussion

We evaluate the performance of the proposed hybrid NDT using data observations acquired by a single UE operating in a closed workspace over a real 5G network.

5.1. Experimental apparatus

The data used to demonstrate the proposed method were collected in a real 5G testbed [30], located at Ericsson's headquarters in Sweden. The testbed is equipped with monitoring and exposure tools that allow for the extraction of network metrics from user equipment, base station, and core network (Fig. 3). The datasets were collected autonomously by a mobile robot in an outdoor area covered by two 5G base stations. The set of measurements Q_i included time, position, user RSRP, and user SINR, with average sampling frequency $f_s = 20$ Hz ($T_s = 0.05$ s).

To validate our methodology, we test two different scenarios against a neural network-based NDT trained as a regression problem with stochastic gradient descent on a fully connected neural network of $n_n = 100$ neurons with ReLu activation function.

5.2. Hybrid NDT approximation with model drift

In the first scenario, the mobile robot scans the workspace in a predefined trajectory. The hybrid NDT converges after approximately 20 s having collected approximately 500 observations and having produced $K = 38$ codevectors $\bar{p}_i$ (respectively regions $\bar{\Sigma}_i$). A regression error event-trigger is identified at time $T_m \approx 30$ s using (24), and the hybrid NDT raises the parameter λ by a factor of 10%, and continues the recursion in (20),(21) using newly collected data. At the same time, the neural network model continues a gradient descent recursion using new data.

In Fig. 4, we illustrate the training behavior of the two approaches. Notice how the proposed hybrid NDT method is faster to converge to low mean-square-error (MSE) values, faster to converge (lower number of observations used), and faster to adapt to the network change. In addition, the classification error related to the identification of the modes S_i , $i = 1, 2$, is depicted in Fig. 5.

Finally, in Fig. 6 we depict the hybrid NDT and the neural network-based RSRP approximation before and after the network change. It is worth noting that the neural network model, even though it presents low regression error, converges to a generic representation that does not favor fast adaptation to local changes. In contrast, the modularity of the hybrid NDT approximation shows the ability to promote faster adaptation and can be used to represent and interpret spatial properties of the network.

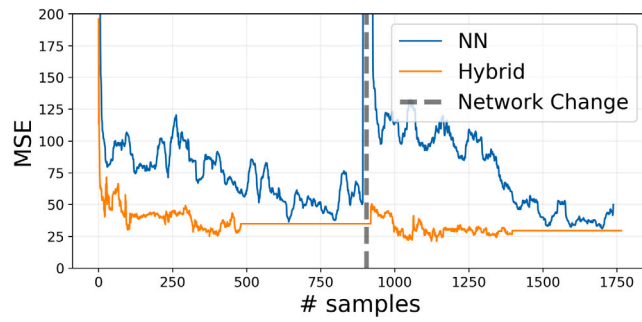


Fig. 4. Comparison of the training behavior of the proposed hybrid NDT with a neural network regressor.

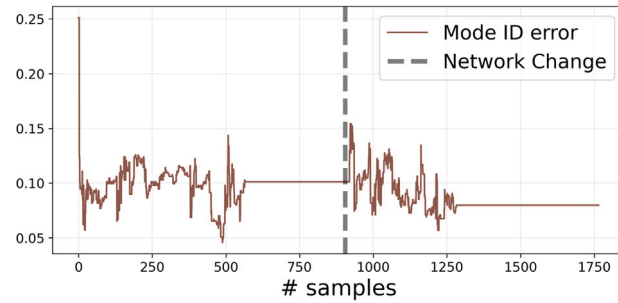
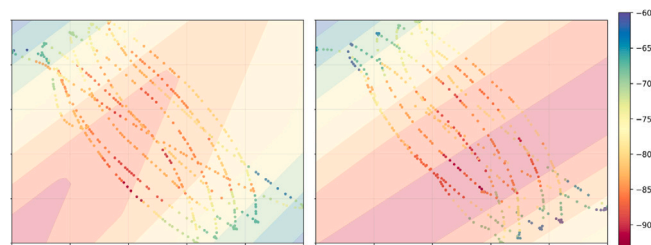
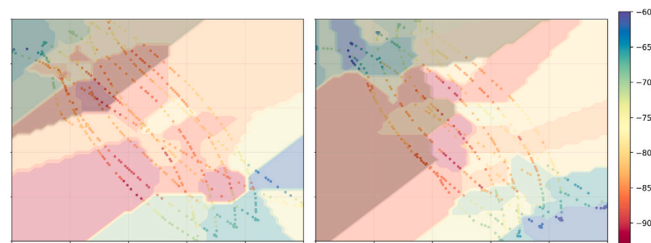


Fig. 5. Mode identification error of the hybrid NDT.



(a) Neural network approximation.



(b) Hybrid NDT approximation.

(b) Hybrid NDT approximation.

Fig. 6. RSRP approximation before (left) and after the network change (right). Shaded areas represent the estimation of mode S_1 . Data samples are also depicted.

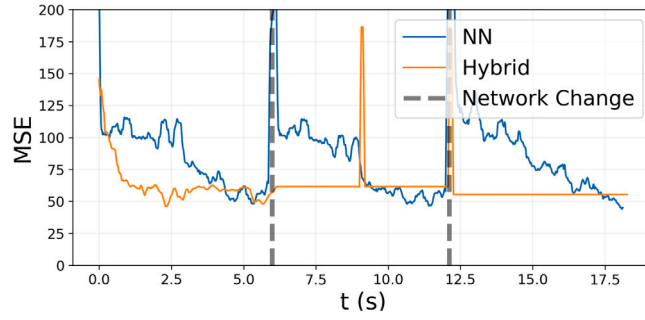


Fig. 7. SINR approximation error during cell malfunction. The hybrid NDT model is not retrained. Instead the term $\Delta_i^{(k)}$ in (11) is used. The spikes correspond to the end of the predefined time duration T in (27).

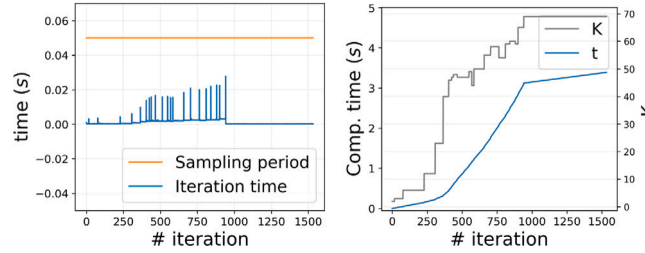


Fig. 8. Iteration and cumulative computation time and their connection to K . Real-time operation is feasible.

5.3. Hybrid NDT adaptation on cell malfunction

In the second scenario, we simulate a temporary cell malfunction by cutting down the user SINR measurements for users connected to cell I_1 to zero for approximately $T_0 = 6$ seconds. In Fig. 7 we show the training behavior of the proposed hybrid NDT against a neural network-based NDT in estimating the SINR component.

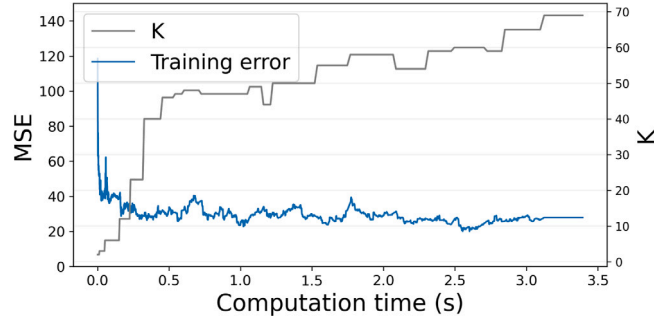
Notice that the hybrid NDT is only training within the first 6 s. When the cell malfunction takes place, the $\Delta_i^{(k)}$ term in (11), (27) is used. In particular, all $\bar{Q}_j$ terms associated with the regions $\bar{\mathcal{S}}_j$ that make up S_1 (i.e., $\bar{c}_j = 1$) are canceled in (11), and $N_1(t, x)$ converges to zero as the solution of a stable linear time-invariant differential equation (with respect to time). In this experiment, a time window duration $T = 3$ s was chosen. For this reason, a spike appears temporarily in the training error at $t \approx 9$ s. At that point, a cell-specific error event-trigger of the form (26) gets activated again.

After the cell malfunction period is over, the model $N_1(t, x)$ converges back to its initially leaned $\bar{Q}_j$ values. That is to say, no more training takes place after $t \approx 6$ s. This is in contrast to the neural network-based NDT model that needs retraining from scratch after any observed change in the network, as shown in Fig. 7.

5.4. Complexity and extensions

The complexity of the spatial characteristics of the network properties can affect the complexity of the algorithm by requiring an increased number of clusters K . Although the number of users in the workspace does not affect the network complexity (it only affects the source of the observations), the number of cells may do so, increasing the time complexity of the algorithm which scales quadratically with K , as shown in (23). In Figs. 8, 9, we present the computational time for the training update iterations, which are completed in a fraction of the available time (T_s) for real-time operation, and how the training error and the cumulative computation time scale with respect to K .

To mitigate this complexity issue in extreme cases, we provide two possible extensions of the proposed approach. The first extension is to impose a tree structure in the construction of the partition of the workspace W , following the approach introduced in [22], reducing the complexity of the algorithm so that it does not scale quadratically with K . The second extension is to try to reduce the number of clusters, extending the proposed method beyond the use of constant local models. More complex learning models can be trained within each cluster $\bar{\mathcal{S}}_i$, using principles from two-timescale stochastic approximation theory as shown in [23]. In that case, the expressive capabilities of the local models can capture high-frequency spatial phenomena like fast fading or complex urban multi-path environments without overly increasing the number of clusters K . It is worth noting that the complexity trade-off that arises between the proposed approaches and the discussed extensions depends highly on the network at hand.

Fig. 9. Training curve with respect to K .

5.5. Towards communication-aware motion planning

Communication-aware motion planning has been gaining significant interest within the communications and robotics communities. The objective is to design optimal robot trajectories such that the connectivity to a base station is maintained and optimized [8,9,31]. In particular, the problem of network-constrained motion planning for n agents can be formulated as an optimal control problem of the form:

Problem 2 (Communication-Aware Optimal Control).

$$\begin{aligned} \min_{\{u_i\}} \quad & \sum_{i=1}^n \int_0^T J_i(t, x_i, u_i) dt + A_i(T) \\ \text{s.t.} \quad & \dot{x}_i = f_i(x_i, u_i), \quad i = 1, \dots, n \\ & x_i(t) \in W_S, \quad \forall t, \quad i = 1, \dots, n \\ & t \in [0, T], \end{aligned} \quad (28)$$

where f_i represent agent dynamics, W_S is a safe (e.g., obstacle-free) space of the workspace, A_i represent terminal costs, and the objective functions J_i are given by:

$$\begin{aligned} J_i(t, x_i, u_i) = & w_i^1 L(t, x_i, u_i) + w_i^2 \phi^c(Q(t, x_i)) \\ & + w_i^3 \phi^h\left(\frac{\partial}{\partial t} \tilde{c}(t, x_i)\right). \end{aligned} \quad (29)$$

The weights $\{w_i^1, w_i^2, w_i^3\}$ are non-negative design parameters that control the trade-off of the three terms in (29) such that $w_i^1 + w_i^2 + w_i^3 = 1$. In particular, L represents a standard optimal control cost function, the scalar function ϕ^c provides a transformation that quantifies a cost related to the communication properties Q (e.g., SINR, latency, throughput), and the term ϕ^h (e.g., $\phi^h(\cdot) = \|\cdot\|^2$) quantifies the cost associated with handover delays with $\tilde{c}(t, x)$ being a filtered signal representing cell association given by the solution of the differential equation:

$$\frac{\partial}{\partial t} \tilde{c}(t, x) = -\gamma_c \tilde{c}(t, x) + c(t, x), \quad t \in \mathbb{R}_+, \quad x \in W, \quad (30)$$

for some $\gamma_c > 0$, and $c(0, x)$, $x \in W$.

While the optimal control problem (28) is provided in continuous time and space, the solution is usually approximated using temporal (and/or spatial) discretization methods. The continuous formulation is supported by the fact that the discretization schemes used by the controller do not necessarily coincide with the ones used in the NDT approximation.

In particular, the spatial modularity of the proposed NDT model can be exploited for the construction of adaptive spatial discretization schemes for the controller that can lead to more efficient multi-agent communication-aware motion planning algorithms.

6. Conclusion

In this work, we studied the identification of an NDT as a hybrid system that models the communication quality properties of a multi-cell, dynamically changing wireless network over a workspace populated with multiple UEs. We explored the advantages of a hybrid NDT, where the discrete modes are associated with areas of the workspace with similar network characteristics, and proposed an annealing optimization-based learning algorithm, driven by online data measurements collected by the UEs, to identify and continuously improve it. Finally, simulations with real experimental data showed that the proposed approach can yield higher memory and computational efficiency, lower data consumption, and increased adaptation speed with respect to network changes. The spatial modularity of the proposed NDT model can be exploited for the construction of efficient motion planning methods for communication-aware control applications.

CRediT authorship contribution statement

Christos Mavridis: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Fernando S. Barbosa:** Writing – original draft, Supervision, Resources, Methodology. **Hamed Farhadi:** Writing – original draft, Supervision. **Karl H. Johansson:** Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Christos Mavridis reports financial support was provided by Swedish Foundation for Strategic Research (SSF) grant IPD23-0019. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] P. Öhlén, C. Johnston, H. Olofsson, S. Terrill, F. Chernogorov, Network digital twins – outlook and opportunities, *Ericsson Technol. Rev.* 2022 (12) (2022) 2–11.
- [2] M.A. Uusitalo, H. Farhadi, M. Boldi, M. Ericson, G. Fettweis, B.M. Khorsandi, I.L. Pavón, M. Mueck, A. Nimr, A. Pärssinen, B. Richerzhagen, P. Rugeland, D. Sabella, H. Wynneersch, Toward 6G — Hexa-X project's key findings, *IEEE Commun. Mag.* 63 (5) (2025) 142–148.
- [3] O. Stenhammar, S. Sorrentino, F. Chernogorov, R. Blasco, G. Fodor, C. Fischione, Real-time network digital twin: Experimental setup for proof of concept, *IEEE Wirel. Commun.* 32 (3) (2025) 114–121.
- [4] X. Lin, L. Kundu, C. Dick, E. Obiodu, T. Mostak, M. Flaxman, 6G digital twin networks: From theory to practice, *IEEE Commun. Mag.* 61 (11) (2023) 72–78.
- [5] A. Jolly, B. Akgun, B. Sachdev, D. Ravichandran, M. Jayabalan, R. Amiri, V. Jain, V. Chande, Y. Chen, R. Acharya, C. Tiwari, C. Woodahl, S. Yerramalli, A. Kannan, X. Zhang, A. Kumar, H. Hong, J. Boyd, R. Prakash, S.B. Mummana, S. Govindappa, J. Wilson, Digital twin powered next generation wireless networks: Construction, validation, and applications, in: 2024 IEEE 100th Vehicular Technology Conference, VTC2024-Fall, 2024, pp. 1–7.
- [6] S. Mihai, M. Yaqoob, D.V. Hung, W. Davis, P. Towakel, M. Raza, M. Karamanoglu, B. Barn, D. Shetve, R.V. Prasad, H. Venkataraman, R. Trestian, H.X. Nguyen, Digital twins: A survey on enabling technologies, challenges, trends and future prospects, *IEEE Commun. Surv. Tutor.* 24 (4) (2022) 2255–2291.
- [7] J.N. Al-Karaki, A.E. Kamal, Routing techniques in wireless sensor networks: a survey, *IEEE Wirel. Commun.* 11 (6) (2004) 6–28.
- [8] A. Muralidharan, Y. Mostofi, Communication-aware robotics: Exploiting motion for communication, *Annu. Rev. Control. Robot. Auton. Syst.* 4 (2021) 115–139.
- [9] D.B. Licea, M. Bonilla, M. Ghogho, S. Lasaulce, V.S. Varma, Communication-aware energy efficient trajectory planning with limited channel knowledge, *IEEE Trans. Robot.* 36 (2) (2019) 431–442.
- [10] Y. Yan, Y. Mostofi, To go or not to go: On energy-aware and communication-aware robotic operation, *IEEE Trans. Control. Netw. Syst.* 1 (3) (2014) 218–231.
- [11] M. Lindhé, K.H. Johansson, Exploiting multipath fading with a mobile robot, *Int. J. Robot. Res.* 32 (12) (2013) 1363–1380.
- [12] M. Lindhé, K.H. Johansson, Adaptive exploitation of multipath fading for mobile sensors, in: 2010 IEEE International Conference on Robotics and Automation, IEEE, 2010, pp. 1934–1939.
- [13] L. Buşoniu, V.S. Varma, J. Lohéac, A. Codrean, O. Ştefan, I.C. Morărescu, S. Lasaulce, Learning control for transmission and navigation with a mobile robot under unknown communication rates, *Control Eng. Pract.* 100 (2020) 104460.
- [14] A. Gupta, K. Ghanshala, R.C. Joshi, Machine learning classifier approach with Gaussian process, ensemble boosted trees, SVM, and linear regression for 5G signal coverage mapping, *IJMAI* 6 (6) (2021) 156–163.
- [15] A. Ghaffarkhah, Y. Mostofi, Communication-aware motion planning in mobile networks, *IEEE Trans. Autom. Control* 56 (10) (2011) 2478–2485.
- [16] Y. Yan, Y. Mostofi, Co-optimization of communication and motion planning of a robotic operation under resource constraints and in fading environments, *IEEE Trans. Wirel. Commun.* 12 (4) (2013) 1562–1572.
- [17] R. Parasuraman, B.C. Min, P. Ögren, Rapid prediction of network quality in mobile robots, *Ad Hoc Netw.* 138 (2023) 103014.
- [18] J. Fink, A. Ribeiro, V. Kumar, Robust control of mobility and communications in autonomous robot teams, *IEEE Access* 1 (2013) 290–309.
- [19] P.K. Penumarthi, A.Q. Li, J. Banfi, N. Basilico, F. Amigoni, J. O'Kane, I. Rekleitis, S. Nelakuditi, Multirobot exploration for building communication maps with prior from communication models, in: 2017 International Symposium on Multi-Robot and Multi-Agent Systems, MRS, IEEE, 2017, pp. 90–96.
- [20] S. Caccamo, R. Parasuraman, L. Freda, M. Gianni, P. Ögren, RCAMP: A resilient communication-aware motion planner for mobile robots with autonomous repair of wireless connectivity, in: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2017, pp. 2010–2017.
- [21] U. Ali, H. Cai, Y. Mostofi, Y. Wardi, Motion-communication co-optimization with cooperative load transfer in mobile robotics: An optimal control perspective, *IEEE Trans. Control. Netw. Syst.* 6 (2) (2018) 621–632.
- [22] C. Mavridis, J. Baras, Multi-resolution online deterministic annealing: A hierarchical and progressive learning architecture, 2023, [arXiv:2212.08189](https://arxiv.org/abs/2212.08189), URL <https://arxiv.org/abs/2212.08189>.
- [23] C. Mavridis, K.H. Johansson, Real-time switched system identification with online deterministic annealing, *IEEE Trans. Autom. Control* (2025).
- [24] C. Mavridis, J.S. Baras, Annealing optimization for progressive learning with stochastic approximation, *IEEE Trans. Autom. Control* 68 (5) (2023) 2862–2874.
- [25] M.Z. Chowdhury, M. Shahjalal, S. Ahmed, Y.M. Jang, 6G wireless communication systems: Applications, requirements, technologies, challenges, and research directions, *IEEE Open J. Commun. Soc.* 1 (2020) 957–975.
- [26] E. Dahlman, S. Parkvall, J. Skold, 5G NR: The Next Generation Wireless Access Technology, second ed. Elsevier, Amsterdam, 2020.
- [27] H. Rydén, H. Farhadi, A. Palaos, L. Héviz, D. Sandberg, T. Kvernvik, Next generation mobile networks' enablers: Machine learning-assisted mobility, traffic, and radio channel prediction, *IEEE Commun. Mag.* 61 (10) (2023) 94–98.
- [28] C.N. Mavridis, J.S. Baras, Online deterministic annealing for classification and clustering, *IEEE Trans. Neural Netw. Learn. Syst.* 34 (10) (2023) 7125–7134.

- [29] C. Mavridis, E. Noorani, J.S. Baras, Risk sensitivity and entropy regularization in prototype-based learning, in: Mediterranean Conference on Control and Automation, 2022, pp. 194–199.
- [30] A. Hernandez, F.S. Barbosa, An end-to-end testbed for communication, compute, and control co-design: The Kista Innovation Park, in: International Conference on Computer Safety, Reliability, and Security, Springer, 2025, pp. 5–16.
- [31] M. Lindhé, K.H. Johansson, Communication-aware trajectory tracking, in: 2008 IEEE International Conference on Robotics and Automation, 2008, pp. 1519–1524.